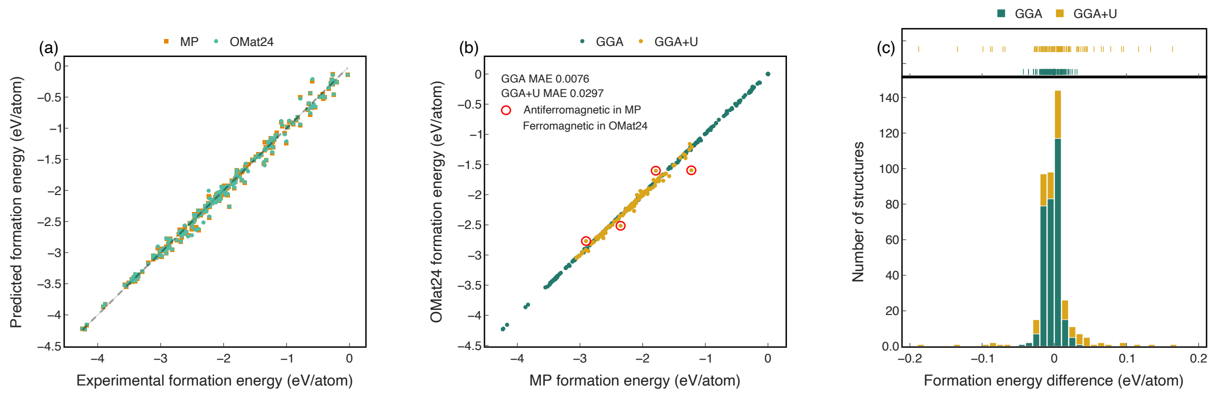

Figure 2 OMat24 and Materials Project DFT formation energy parity. Formation energy computed with Materials Project PBE settings and OMat24 DFT settings for 423 selected compounds from Materials Project. (a) Parity plot of predicted and experimental formation energy. (b) Parity plot between predicted formation energies using OMat24 and MP DFT settings.
(c) Histogram of energy differences. Formation energies are calculated using MP2020 GGA/GGA+U mixing corrections and anion composition corrections. MP corrections are taken directly from MP. OMat24 corrections are fitted following the same procedure using DFT data calculated with OMat24 settings.

magnetic orderings. Addressing magnetic structure in MLIPs remains an active area of research, involving both the generation of balanced and informative training datasets[42] and advancements in model architecture and training strategies[16,43].

Finally, the OMat24 dataset was generated using convergence criteria aligned with Materials Project PBE calculations[19], striking a balance between computational efficiency and accuracy to support large-scale data generation within available compute resources. Later analyses has identified AIMD subsplits exhibiting nonzero drift forces, suggesting that tighter convergence could be beneficial[44]. However, as shown in the Supplementary Table B.1, model performance remains robust when trained on these subsplits and evaluated against calculations with tighter convergence thresholds and zero drift.

2.2 OMat24 model evaluations

We evaluate the performance of MLIPs trained on the OMat24 dataset and compare against models trained on alternative datasets using the popular Matbench-Discovery benchmark[29]. Both eSEN[28] and EquiformerV2[26] trained on the OMat24 dataset surpass previous state-of-the-art models. Most notably, all five of the currently top-performing models, trained by ourselves or others, on the leaderboard are exclusively trained on the OMat24 dataset. This underscores the critical role of large diverse datasets in advancing MLIP performance.

In addition, we investigate the pervasive issue of systematic underprediction of energy, forces, and phonons—referred to as systematic softening[23]. Our analysis shows that this detrimental effect is dramatically reduced or entirely eliminated across diverse model architectures when trained on the OMat24 dataset. This breakthrough highlights the dataset's unique ability to enhance model reliability and generalizability to non-equilibrium regimes.

2.2.1 Model performance on Matbench-Discovery

The Matbench discovery currently evaluates models based on two prediction tasks: (1) a materials discovery task and (2) a thermal conductivity prediction task. The discovery task predicts thermodynamic stability by comparing a material's decomposition energy to competing phases[45–47]. This requires relaxing atomic positions and the crystal cell, which is more computationally expensive with DFT than with MLIPs. Machine Learning Interatomic Potentials (MLIPs) offer much faster predictions. The recently included thermal conductivity (κ) prediction task[29,48] requires MLIPs to accurately capture second and third derivatives of the potential energy surface (PES) in order to make accurate predictions of thermal transport.

Table 2 summarizes results for models pre-trained on OMat24 and OC20 (metric definitions in Supplementary Table G.5). Because OMat24 uses different DFT settings than the Matbench Discovery leaderboard, we fine-tuned on MPtrj (OMP) or jointly on sAlex and MPtrj (OAM). Pre-training on OC20—despite not being designed for materials modeling—performs strongly after MPtrj fine-tuning, highlighting the value of large-scale non-equilibrium data diversity. Pre-training on OMat24 yields substantially better results across nearly all metrics, clearly outperforming prior methods. The resulting eSEN and equiformerV2 models set the new best performance on Matbench-Discovery at the time of writing: eSEN achieves an F1 score of 0.925 and an energy-above-hull MAE of 18 meV/atom. The gains from OMat24 are also evident relative to models trained only on MPTrj, with and without denoising (Supplementary Figure F.1).

Table 2 Matbench-Discovery benchmark results of non-compliant models on the unique protoype split. Mean absolute error (MAE) and Root mean squared error (RMSE) are in units of eV/atom.

Model Pretrain Finetune	eSEN-30M OMat MPtrj-sAlex	eqV2-L OMat MPtrj-sAlex	eqV2-M OMat MPtrj-sAlex	eqV2-M OMat MPtrj	eqV2-S OMat MPtrj-sAlex	eqV2-S OMat MPtrj	eqV2-L OC20 MPtrj	eqV2-S OC20 MPtrj
F1 ↑	**0.925**	0.915	0.917	0.909	0.901	0.89	0.86	0.837
DAF ↑	6.069	**6.113**	6.047	5.948	5.902	5.752	5.639	5.392
Precision ↑	0.928	**0.934**	0.923	0.909	0.902	0.879	0.862	0.824
Recall ↑	**0.923**	0.897	0.910	0.909	0.9	0.901	0.858	0.849
Accuracy ↑	0.977	**0.985**	0.975	0.973	0.97	0.966	0.957	0.951
TPR ↑	**0.923**	0.897	0.91	0.909	0.9	0.901	0.858	0.849
FPR ↓	0.013	**0.012**	0.014	0.017	0.018	0.023	0.025	0.033
TNR ↑	0.987	**0.988**	0.986	0.983	0.982	0.977	0.975	0.967
FNR ↓	**0.077**	0.103	0.090	0.091	0.1	0.099	0.142	0.151
MAE ↓	**18**	19	20	21	24	26	29	33
RMSE ↓	**67**	71	72	72	80	81	78	80
R2 ↑	**0.866**	0.852	0.848	0.849	0.811	0.807	0.823	0.810

More importantly, the substantial improvements from models trained using OMat24 extend across architectures beyond eSEN and equiformerV2. Figure 3 displays the mean absolute error for energy above hull and the symmetric relative mean error (SRME) for thermal conductivity, presented in chronological order for models on the Matbench-Discovery leaderboard [29] from its inception to early 2025, six months after the release of the OMat24 preprint https://arxiv.org/abs/2410.12771.

The five top performing models trained using OMat24 achieve unprecedented performance in the discovery tasks, with many achieving energy above hull prediction errors near 20 meV/atom. Additionally, we observe that all *energy conserving* models trained using OMat24 exhibit the lowest thermal conductivity prediction errors and consistently top the overall Matbench-Discovery leaderboard. We note that direct force models, such as equiformerV2 and ORB v2, have been found to perform poorly in phonon prediction tasks [49,50], as evidenced by their high thermal conductivity SRME.

2.2.2 Model evaluation of systematic softening

A systematic underprediction of energy, forces, and phonons has been observed across MLIP architectures trained on smaller datasets of relaxation trajectories, such as MPtrj [23]. This phenomenon, known as systematic softening, is attributed to dataset distribution bias and a lack of non-equilibrium structures.

Building on the work of Deng et al. [23], we quantify systematic softening at three levels based on the order of potential energy derivatives used for prediction.

1. **Zeroth order softening** quantifies the systematic underprediction of energy in non-equilibrium structures, such as those with defects, surfaces, and interfaces. We measure this as the difference between the target and predicted energy per atom. A negative bias indicates zeroth-order softening.
2. **First order softening** quantifies the systematic underprediction of forces in non-equilibrium structures. We assess this using the previously proposed *softening scale* [23], which is the slope of the least squares

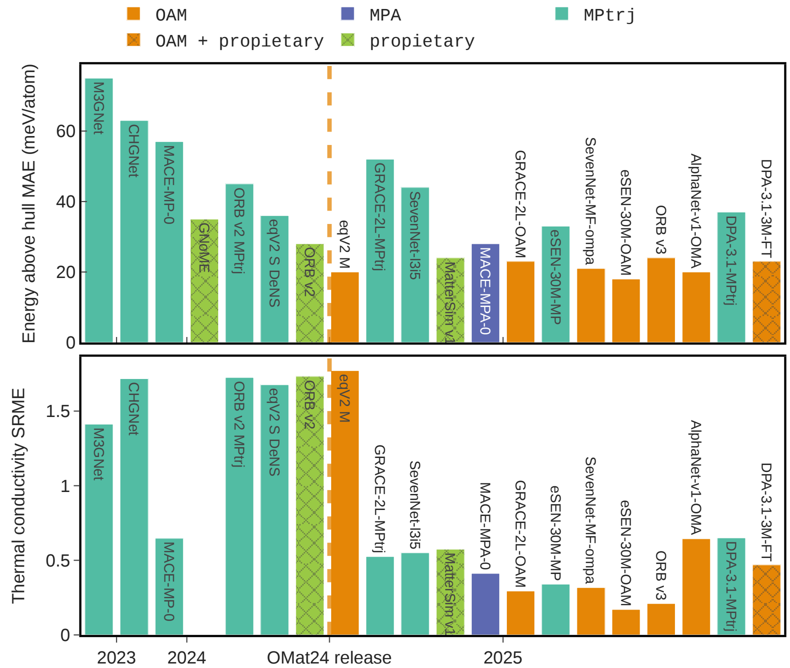

Figure 3 Chronological performance improvements of MLIP models. Energy above hull mean absolute error and thermal conductivity symmetric relative mean error (κ_{SRME}) of models on Matbench-Discovery [29] in chronological release order. The majority top performing models are trained on the OMat24 dataset and finetuned on MPtrj [16] and sAlex [20], labeled as OAM.

fit to the parity plot of target forces versus predicted forces. A bias towards values smaller than one indicates first-order softening.

3. **Second order softening** quantifies the systematic underprediction of phonon vibrational frequencies. We measure this as the ratio between the maximum predicted and maximum target phonon frequency. A bias towards values smaller than one indicates second-order softening.

We evaluate zeroth-order and first-order softening using the WBM high energy states dataset [23], which comprises approximately ten thousand high-temperature MD samples from randomly selected WBM structures [34]. We estimate second-order softening using the PBE MDR phonon dataset [49]. All calculations in these two datsets use the PBE functional and the same pseudopotentials; the MDR phonon calculations differ primarily in their more stringent force relaxation and electronic energy convergence criteria [49]. We note that the PBE MDR phonon dataset excludes materials requiring $+$U corrections per the Materials Project protocol [51].

Figure 4 shows violin plots quantifying (a) zeroth-order, (b) first-order, and (c) second-order systematic softening for models trained on MPtrj only, pretrained on OMat24 and finetuned on MPtrj and sAlex (OAM), and trained on OMat24 only. MACE models trained on MPtrj and MPtrj $+$ sAlex without OMat24 pretraining are included as baselines; results for additional third-party models appear in Supplementary Figure E.1. OAM models show a notable reduction in zeroth- and first-order softening compared to MPtrj-only models, with the eSEN and equiformerV2 architectures nearly eliminating it entirely. The MACE model trained on MPtrj and sAlex only (MPA) also exhibits reduced softening relative to the MPtrj-only model, though to a lesser extent than the OAM models.

The second-order (phonon) softening results in Figure 4a show a regression for the eSEN OAM model, which exhibits more softening than its MPtrj counterpart. In contrast, the GRACE and SevenNet OAM models show moderate reductions in softening (Supplementary Figure E.1). To understand this regression, we examine

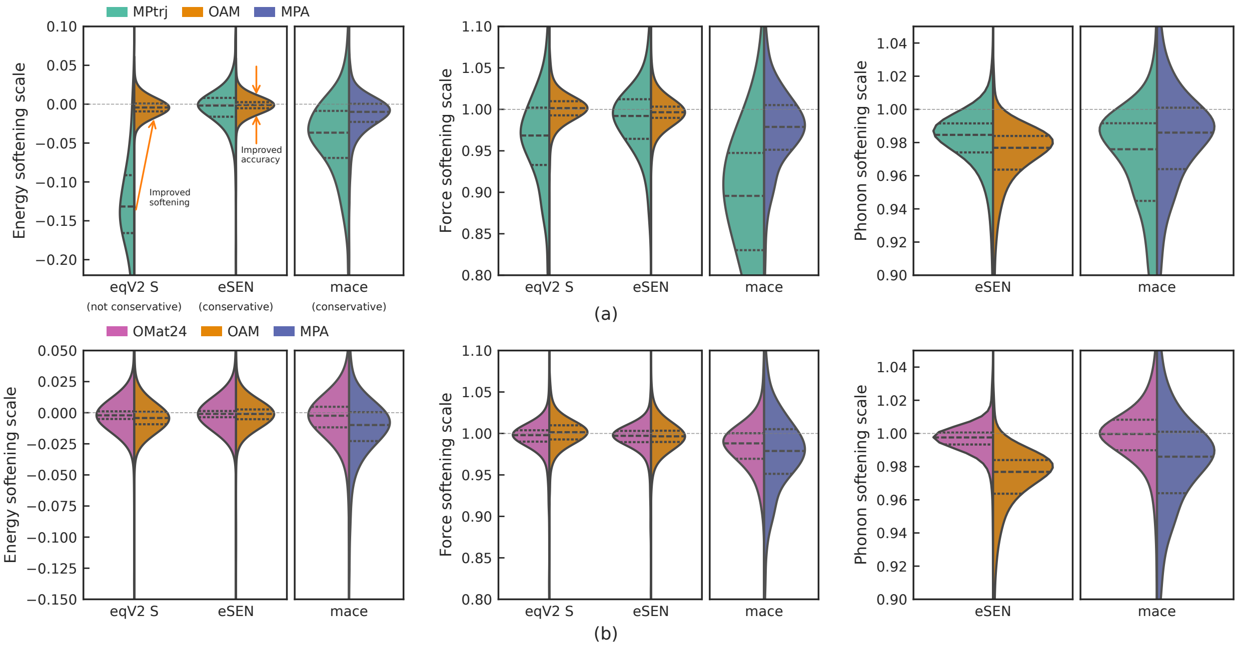

Figure 4 Reduction of systematic softening across models using OMat24 training data Zeroth order (energy), first order (forces) and second order (phonon) systematic softening distribution shifts between (a) MPTrj, OAM (OMat24 + MP/sAlex finetuning) and MPA (MP/sAlex) trained models and (b) OMat24, OAM, and MPA models. Violin plots show the distribution of the energy and force softening scale of model predictions on high energy WBM set [23], and phonon softening of conservative model predictions on the recalculated MDR phonon dataset [49]. Models are labeled *conservative* if forces are predicted as energy gradients, and *not conservative* otherwise.

models trained on OMat24 alone without fine-tuning (Figure 4b). The eSEN OMat24-only model does not exhibit second-order softening, highlighting the benefit of the dataset's diversity and breadth. Indeed, models trained solely on OMat24 show less softening of all orders across all architectures, suggesting that fine-tuning with relaxation data reintroduces some systematic softening.

3 Discussion

The OMat24 dataset is a major step forward for materials science, providing a large, diverse set of PBE-DFT structures for training ML interatomic potentials. Its breadth enables models to capture complex material behavior and considerably improve property prediction accuracy. While OMat24 samples a broad spectrum of non-equilibrium configurations, it does not explicitly include transition-state frames or highly off-equilibrium regions such as bond-breaking or large coordination changes. Most existing benchmarks focus on equilibrium or near-equilibrium properties; nevertheless, models trained on OMat24 have shown strong extrapolation capabilities, successfully predicting properties for states not directly represented in the dataset—including defects, grain boundaries, adsorption, and cleavage energies [52–54]. Ongoing development of such benchmarks remains essential for identifying model limitations and guiding future dataset generation efforts.

Furthermore, discrepancies between the PBE functional and experimental results can limit the reliability of predictions for real-world applications. Users should be mindful of the inherent limitations of the PBE(+U) functional and the dataset's emphasis on bulk, defect-free structures. In particular, PBE and related GGA functionals often struggle to accurately capture strong on-site Coulomb interactions in d and f orbitals, which may lead to deviations from experimental observations. [55].

Developing training sets and benchmarks using more accurate DFT functionals such as SCAN and r2SCAN could help in starting to overcome these challenges [56]. As of release v2022.10.28 [57], the Materials Project began including r2SCAN calculations for a portion of its dataset. Additionaly, MatPES, the first r2SCAN MLIP training dataset containing 400K structures[58] was released shortly after the release of OMat24. Large-scale datasets similar to OMat24 but with SCAN, r2SCAN or higher fidelity functionals along with additional physically motivated sampling strategies, such as using active learning cycles, molecular dynamics, and metadynamics to efficiently sample and optimize configurations beyond local minima are obvious next steps.

Efficiently building models with higher-fidelity methods is also possible through multi-fidelity learning, delta learning, or fine-tuning. These approaches can leverage the extensive low-level force data available in OMat24, supplemented by a smaller set of high-level energies, allowing MLIPs to achieve hybrid or even CCSD(T)-level accuracy with reduced computational cost[59–61]. By using OMat24's elemental diversity and non-equilibrium sampling as a pretraining foundation, researchers can accelerate the development of high-accuracy potentials for applications where functional errors or surface and defect physics are critical.

We anticipate that the release of OMat24 will facilitate more data-efficient modeling efforts. For instance, the MatPES project demonstrated the value of maximizing information content in smaller datasets[18], and recent work has focused on effective sampling and dataset compression strategies[62,63]. OMat24 provides a robust platform to further investigate sampling and compression techniques aimed at identifying non-redundant physical information, ultimately enabling the development of state-of-the-art models trained on minimal data.

The OMat24 dataset has quickly become a pivotal resource in developing highly accurate MLIPs, greatly reducing systematic softening biases. Its comprehensive coverage enable the creation of models that not only excel in predictive tasks but also provide a reliable foundation for future advancements in materials science. Similar to how this work builds upon open research efforts, like the Materials Project[19] and the Alexandria datasets[20], we hope this work enables the research comunity to drive further advancements in the capabilities of machine learning models for materials science.

4 Methods

4.1 OMat24 structure generation and sampling

Input structure generation for the OMat24 dataset consists of three different processes intended to obtain diverse non-equilibrium structures: Boltzmann sampling of rattled (random Gaussian perturbations of atomic positions) structures, ab initio molecular dynamics (AIMD), and relaxations of rattled structures. These methods were used to increase the diversity of sampled configurations, similar to prior large dataset efforts[10,24].

In all three approaches, initial structures were obtained by randomly sampling the relaxed structures in the Alexandria PBE bulk materials dataset (3D compounds)[20]. The Alexandria dataset was chosen as a starting point because it was the largest openly available DFT dataset of equilibrium and near equilibrium structures ($\sim$4.5 million materials). By randomly sampling relaxed structures from the Alexandria dataset we are able to cover a wide elemental composition diversity. Additionally, using Alexandria relaxed structures as a starting point prevents generating structures too far from equilibrium that could result in DFT convergence errors, or unphysical starting configurations. The details for the three processes are as follows:

Rattled Boltzmann sampling: For each randomly sampled Alexandria structure we generated 500 candidate non-equilibrium structures by scaling the unit cell to contain at least 10 atoms. Atomic positions were then rattled with displacements sampled from a Gaussian distribution of $\mu = 0$Å and $\sigma = 0.5$Å. Unit cells were also deformed isotropically and anisotropically from a Gaussian distribution of $\mu = 1\%$ and $\sigma = 5\%$. For each set of 500 candidate structures we selected 5 of them from a Boltzmann-like distribution based on total energies predicted by an EquiformerV2 model trained on the MPtrj dataset. The sampling procedure was done using three different sampling temperatures: 300K, 500K and 1000K.

Ab-Initio Molecular Dynamics (AIMD): Short-length (50 ionic steps) ab-initio molecular dynamics were carried out starting from randomly sampled relaxed structures in the Alexandria dataset. Structures were recorded from constant temperature and volume (NVT), and constant temperature and pressure (NPT) AIMD trajectories at temperatures of 1000K and 3000K. Unit cells were scaled to contain at least 50 atoms for structures sampled at 3000K.

Rattled relaxation: Relaxed structures from Alexandria were selected at random, rattled (both atomic positions and unit cell), and re-relaxed. Atomic displacements were sampled from a Gaussian distribution